HU Collection

Test of Mathematics for University Admission

Mock Set B Paper 1

Applications of Mathematical Knowledge

Time allowed:	**75 minutes**
Questions:	20 multiple choice
Calculators:	not permitted
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Date written:	18 September 2026

Name: _____

Date sat: _____

Score: _____ / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Unless a question says otherwise, all angles are in radians and all logarithms may be taken to any fixed base. Record your answers as letters on the grid below before consulting the solutions.

Difficulty note. This paper contains no routine questions. Every item is pitched at the level of the final third of a real TMUA paper, and most rely on a method or trap that a standard school course does not teach. Finishing in 75 minutes is meant to be difficult; score by what you get right, not by what you finish.

Q	1	2	3	4	5	6	7	8	9	10
Ans										
Q	11	12	13	14	15	16	17	18	19	20
Ans										

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Question 1

Find the exact value of $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$.

A $\frac{1}{16}$

B $\frac{1}{8}$

C $\frac{1}{4}$

D $\frac{\sqrt{7}}{8}$

E $-\frac{1}{8}$

F $\frac{1}{2}$

Question 2

Evaluate

$$\int_{-1}^{5} \left| x^3 - 6x^2 + 12x - 9 \right| dx.$$

A -6

B 6

C 24

D 36

E 42

F 48

Question 3

A polynomial p of degree 4 satisfies $p(k) = 2^k$ for $k = 0, 1, 2, 3, 4$. Find $p(5)$.

A 16

B 30

C 31

D 32

E 33

F 64

Question 4

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Let a, b and c be the three real roots of $x^3 - 3x + 1 = 0$. Find the value of

$$(a^2 - 2)(b^2 - 2)(c^2 - 2).$$

A -3

B -1

C 0

D 1

E 3

F 9

Question 5

Positive real numbers x and y satisfy $x + y = 1$. Find the minimum value of

$$\left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2.$$

A 8

B $\frac{25}{4}$

C $\frac{25}{2}$

D 13

E 15

F 25

Question 6

Let M and m be the maximum and minimum values of

$$(\sin x + 2\cos x)(2\sin x + \cos x)$$

over all real x. Find Mm.

A $-\frac{9}{4}$

B $-\frac{5}{2}$

C -1

D 0

E $\frac{9}{4}$

F $\frac{81}{4}$

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Question 7

Find the value of

$$\sum_{n=1}^{\infty} \frac{n+2}{n(n+1)\,2^n}.$$

A $\frac{1}{2}$

B $\frac{3}{4}$

C 1

D $\frac{5}{4}$

E $\frac{3}{2}$

F 2

Question 8

Find the sum of all solutions of

$$\sin x \cos x = \sin x + \cos x - 1$$

in the range $0° \leq x \leq 360°$.

A $90°$

B $180°$

C $450°$

D $540°$

E $720°$

F $900°$

Question 9

A regular heptagon has side length 1. Its diagonals take exactly two distinct lengths, $p < q$. Find the value of $\frac{1}{p} + \frac{1}{q}$.

A $\frac{1}{2}$

B $\frac{\sqrt{2}}{2}$

C 1

D $\frac{\sqrt{7}}{2}$

E $\frac{3}{2}$

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F 2

Question 10

The roots of $x^2 - 2x - 1 = 0$ are α and β. Find the value of

$$\frac{1}{\alpha^5} + \frac{1}{\beta^5}.$$

A -82

B -41

C $-\frac{1}{82}$

D $\frac{1}{82}$

E 41

F 82

Question 11

There are exactly two straight lines that are tangent to both of the curves

$$y = x^2 \qquad \text{and} \qquad y = -x^2 + 6x - 10.$$

Find the product of the gradients of these two lines.

A -4

B -2

C -1

D 1

E 2

F 4

Question 12

Find the complete set of real values of x for which the infinite series

$$1 + (x-1) + (x-1)^2 + (x-1)^3 + \cdots$$

converges and has sum equal to $\frac{1}{2-x}$.

A all real x with $x \neq 2$

B $x < 2$

C $0 < x < 2$

D $0 \leq x < 2$

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E $1 < x < 2$

F $x > 0$

Question 13

How many chords of the circle $x^2 + y^2 = 25$ have length 6 and pass through the point $(3, 1)$?

A 0

B 1

C 2

D 3

E 4

F infinitely many

Question 14

A sequence is defined by $a_1 = 3$ and

$$a_{n+1} = \frac{a_n - 1}{a_n + 1} \qquad (n \geq 1).$$

Find the value of the product $a_1 a_2 a_3 \cdots a_{2026}$.

A -2

B $-\frac{1}{2}$

C 1

D $\frac{3}{2}$

E 3

F $\frac{9}{4}$

Question 15

Find the complete set of real values of k for which the equation

$$4^x + 4^{-x} = k\left(2^x + 2^{-x}\right) - 4$$

has exactly two real solutions in x.

A $k \geq 2\sqrt{2}$

B $k > 2\sqrt{2}$

C $k \geq 3$

D $k > 3$

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E $|k| > 3$

F $k > 2$

Question 16

The curve $y = x^4 - 4x^3 + ax^2$, where a is a real constant, has exactly one stationary point. Find the complete set of possible values of a.

A $a > \frac{9}{2}$

B $a \geq \frac{9}{2}$

C $a < \frac{9}{2}$

D $0 < a < \frac{9}{2}$

E $a > 0$

F $a \geq 0$

Question 17

Evaluate

$$\int_{-2}^{2} (x+2)\sqrt{4-x^2}\,dx.$$

A 0

B 8

C 2π

D 4π

E 8π

F 16

Question 18

The points $A(1,2)$ and $B(5,2)$ are fixed, and C is a point such that $\angle ACB = 45°$. Find the maximum possible area of triangle ABC.

A $2 + 2\sqrt{2}$

B $4\sqrt{2}$

C $4 + 2\sqrt{2}$

D $4 + 4\sqrt{2}$

E 8

F $8\sqrt{2}$

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Question 19

A function f satisfies

$$\int_0^2 f(2x)\,dx = 3, \qquad \int_1^3 f(x+1)\,dx = 5, \qquad \int_0^1 f(3-3x)\,dx = 2.$$

Find the value of $\displaystyle\int_3^4 f(x)\,dx$.

A -1

B 0

C 1

D 2

E 5

F 6

G cannot be determined from the information given

Question 20

Let $f(x) = x^3 + x$, and let f^{-1} denote the inverse function of f. Find the value of

$$\int_0^{10} f^{-1}(x)\,dx.$$

A 6

B 10

C 14

D 16

E 20

F 34

END OF PAPER

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